

Math 70

Week 7

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1 Problems

Example 1.1. Let $Y = X^2$, where $X \sim \text{Unif}(-1, 1)$. Find the mean and variance of Y

Example 1.2. Let X and Y be uniformly distributed independent RVs on interval $(0, 1)$. Define $Z = \min(X, Y)$. Find the mean and variance of Z

Example 1.3. Let X and Y be uniformly distributed independent RVs on interval $(0, 1)$. Define $Z = \max(X, Y)$. Find the mean and variance of Z

Example 1.4. (Old MT problem) Let X be a Binomial random variable with $E[X] = 30$ and $\text{Var}(X) = 21$. Calculate $F(4) - F(3)$, where F is the CDF of X .